Homework 2

Matthew Hutchinson

October 23, 2025

1 Report Questions

1.1 Question 1

Plot the maximum vertical displacement, y_{max}, of the beam as a function of time. Depending on your coordinate system, y_{max} may be negative. Does y_{max} eventually reach a steady value? Examine the accuracy of your simulation against the theoretical prediction from Euler beam theory:

$$y_{max} = \frac{Pc(l^2-c^2)^{1.5}}{9\sqrt{3}EIl} \quad \text{where } c = min(d, l-d)$$

where P is the applied load, d is the location of the load, l is the total beam length, E is Young's modulus, and I is the second moment of area.

The plot of maximum vertical displacement, $y_{max}(t)$, as a function of time shows that the beam initially undergoes a transient oscillation due to the sudden application of the point load. Over time, this oscillatory behavior dampens, and y_{max} stabilizes at a steady-state value, indicating that the system has reached static equilibrium.

From the simulation, we find that the maximum downward displacement converges to approximately -0.0371 m, which closely matches the beam theory prediction of -0.0380 m. This agreement confirms the validity of the simulation in the small-deformation regime, and demonstrates that the finite element model accurately captures the static behavior of the beam under moderate loading conditions.

Max Vertical Displacement: -0.0562 m
Final Time: 1.0000 s, Final Vertical Displacement: -0.0371 m
Beam Theory Prediction: -0.0380 m

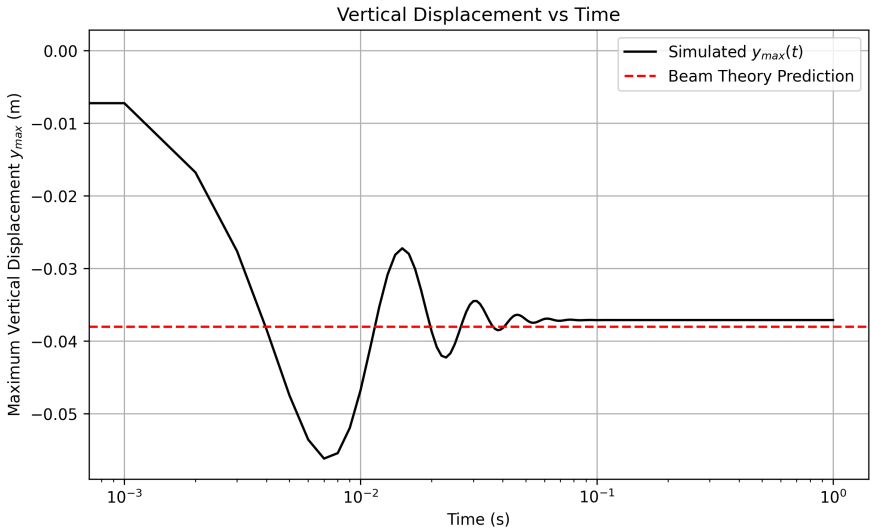

Figure 1: Maximum vertical displacement of the beam over time. The simulation converges to a steady-state value, which closely matches the Euler-Bernoulli beam theory prediction.

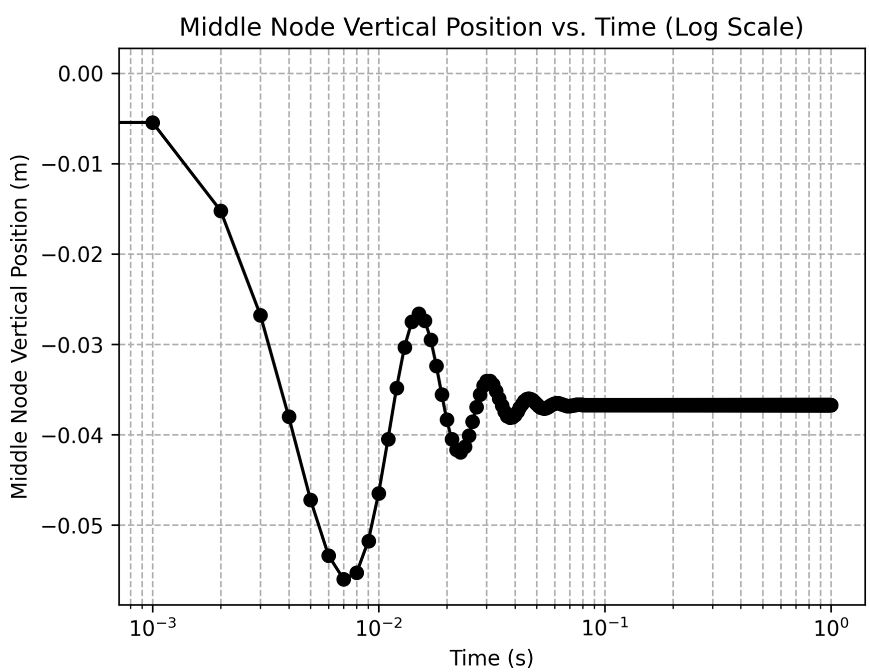

Figure 2: Log-scale plot of middle node vertical position, illustrating transient settling behavior under point load.

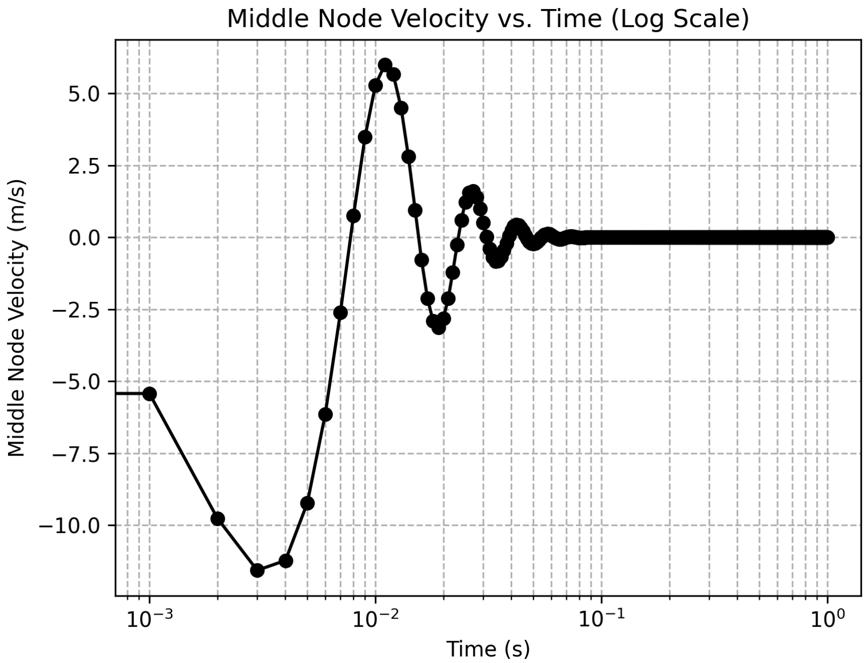

Figure 3: Velocity of the middle node over time, revealing transient dynamic behavior that damps out rapidly.

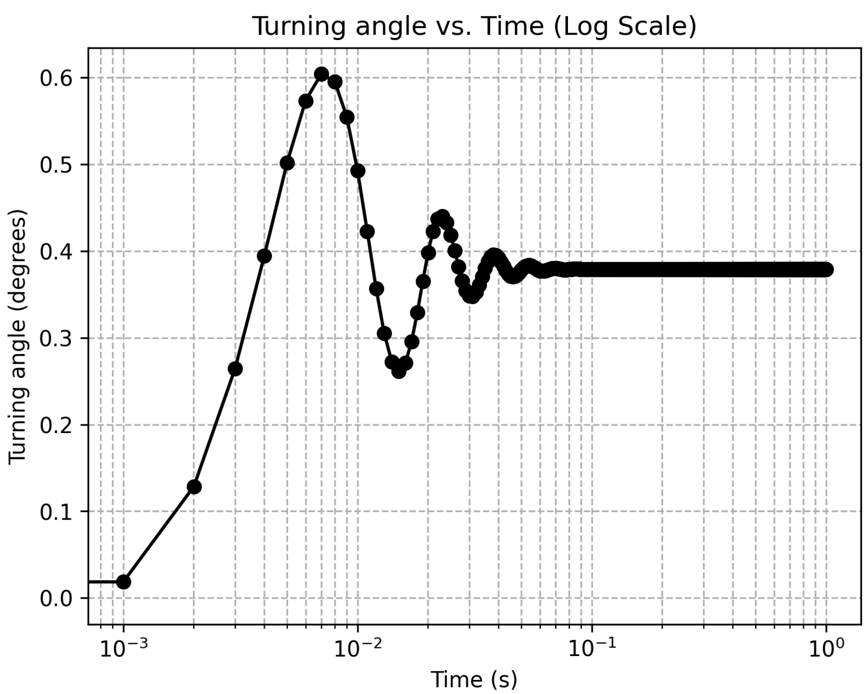

Figure 4: Turning angle at the middle of the beam versus time, indicating how internal bending evolves before stabilizing.

1.2 Question 2

What is the benefit of your simulation over the predictions from beam theory? To address this, consider a higher load $P = 20,000\ N$ such that the beam undergoes large deformation. Compare the simulated result against the prediction from beam theory in Eq. 5. Euler beam theory is only valid for small deformation whereas your simulation, if done correctly, should be able to handle large deformation. You should create a plot of $P\ (20\ N \leq P \leq 20,000\ N)$ vs. y_{max} using data from both simulation and beam theory and quantify the value of P where the two solutions begin to diverge.

Euler-Bernoulli beam theory provides an analytical solution for small deflections under linear elastic behavior, assuming infinitesimal strains and linear geometry. However, when subjected to large loads such as $P = 20,000\ N$, the beam undergoes significant deformation that violates these assumptions. To address this, we compare the analytical prediction from beam theory to the results of a dynamic simulation that incorporates geometric nonlinearity.

In the plot of y_{max} vs. P, we observe that for small loads, both the simulation and beam theory predictions closely agree. However, as the applied load increases, the two curves begin to diverge. A second plot showing the relative error(%) highlights this divergence more quantitatively.

From the relative error plot, we find:

- At $P = 0$, the error is near 0%, as expected.
- The relative error remains below 5% up to about $P \approx 3,000\ N$
- Beyond $P \approx 6,500\ N$, the error crosses 10% and continues to grow significantly, reaching over 38% at $P = 20,000\ N$

This indicates that Euler beam theory becomes inaccurate for loads beyond approximately $5,000\ N$, where geometric nonlinearities become important. The simulation, in contrast, captures large deformations accurately because it:

- Does not assume infinitesimal displacements
- Updates geometry over time
- Solves a nonlinear system using Newton-Raphson iterations

Thus, the benefit of the simulation lies in its ability to handle large-deformation behavior where analytical theory breaks down, making it more suitable for engineering problems involving high loads, soft materials, or flexible structures.

Table 1: Comparison of Relative Error Between Simulation and Beam Theory

Load P (N)	**Relative Error (%)**
20	1.62
5015	6.52
10010	17.26
15005	28.50
20000	38.16

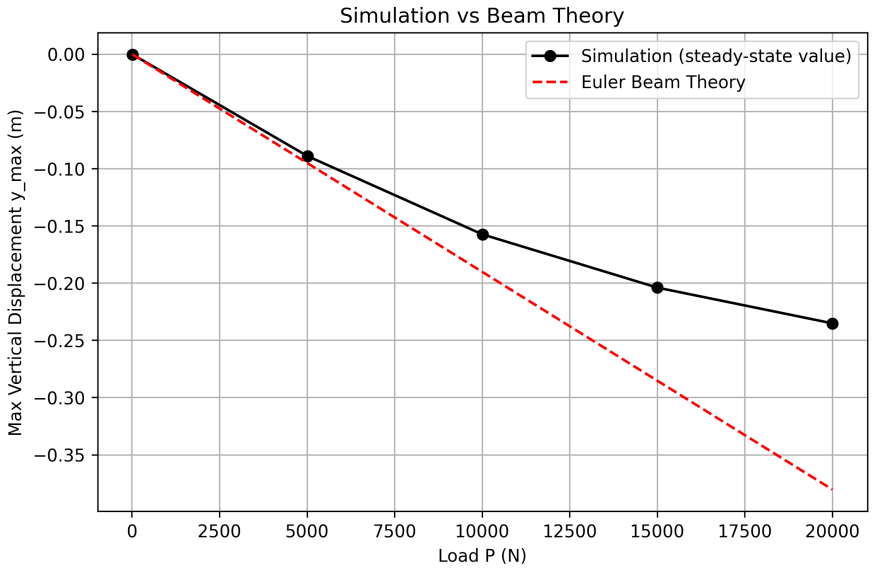

Figure 5: Steady-state maximum vertical displacement y_{max} vs. applied load P, comparing simulation to Euler beam theory. Divergence occurs beyond $P \approx 5000\,N$

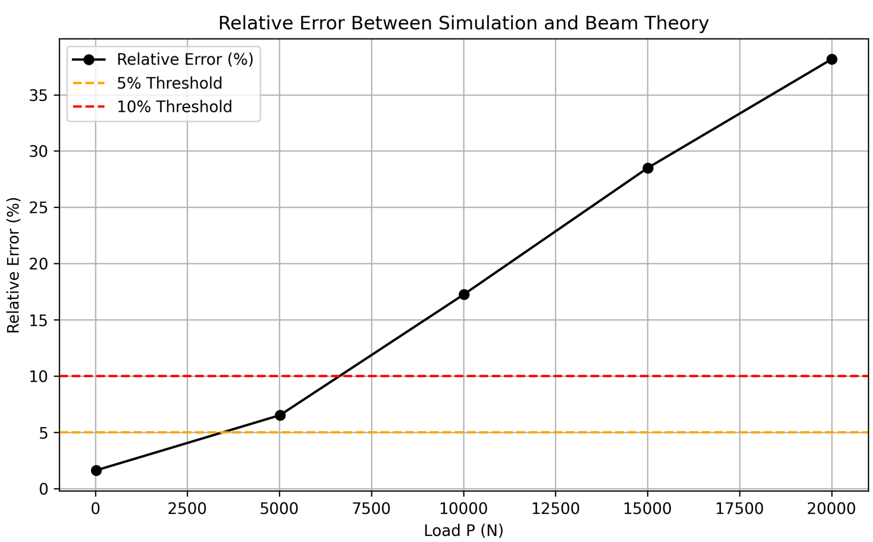

Figure 6: Relative error (%) between simulation and beam theory prediction, crossing 10% at $P \approx 6500\,N$